Q01

A 45-year-old man with a duodenal ulcer is prescribed omeprazole. Which of the following best describes its mechanism of action?

A Irreversibly inhibits H+/K+ ATPase in parietal cells

B Blocks histamine H2 receptors on parietal cells

C Neutralizes gastric acid by increasing pH directly

D Forms a protective layer over ulcer base

Q02

Which of the following drugs is the prototype of a potassium-competitive acid blocker (P-CAB)?

A Omeprazole

B Pantoprazole

C Misoprostol

D Vonoprazan

Q03

A 32-year-old woman with chronic arthritis is prescribed misoprostol for ulcer prevention. She develops abdominal cramping and diarrhea. What other contraindication should be considered?

A Diabetes

B Pregnancy

C Hypertension

D Peptic ulcer

Q04

A 62-year-old patient on clopidogrel for ischemic heart disease develops GERD. Which acid-suppressing agent should be avoided due to inhibition of CYP2C19, reducing clopidogrel activation?

A Ranitidine

B Omeprazole

C Pirenzepine

D Famotidine

Q05

How do P-CABs differ from PPIs in their mechanism of action?

A P-CABs are prodrugs that require activation by acid.

B P-CABs bind irreversibly to the HK-ATPase pump.

C P-CABs directly and reversibly inhibit the K-binding site on the gastric proton pump.

D P-CABs inhibit acid production by blocking histamine H2 receptors.

Q06

A 40-year-old male came to the emergency department with severe burning chest pain radiating into his neck. His electrocardiogram was normal and test for troponin was negative. A diagnosis of GERD was made and he was sent home with a prescription for a drug that inhibits stomach acid. Which of the following is a drug that irreversibly inhibits the H+/K+ ATPase in the parietal cells?

A Cimetidine

B Diphenoxylate

C Esomeprazole

D Metoclopramide

Q07

When is the best time to take a PPI for optimal effectiveness?

A Right before bed

B During a meal

C 30–60 minutes before the first meal of the day

D Immediately after a meal

Q08

Which of the following is a long-term risk associated with overprescribing PPIs?

A Increased blood pressure.

B Increased bone fracture risk (hypomagnesemia/malabsorption).

C Increased stomach motility.

D Kidney stones